CSE2023 ASSIGNMENT #5 (WILL NOT BE GRADED)

1) Give an explicit formula for s_n where $s_0 = 3$, $s_1 = -3$ and $s_n = s_{n-1} + 2s_{n-2}$ for $n \geq 2$.

2) Consider the sequence s_n where $s_0 = 2$, $s_1 = 1$ and $s_n = s_{n-1} + s_{n-2}$ for $n \geq 2$. Give an explicit formula for s_n.

3) Give an explicit formula for s_n.
   a) $s_0 = 0$ and $s_n = 5s_{n-1}$ for $n \geq 1$.
   b) $s_0 = 0$, $s_1 = 0$, $s_2 = 0$ and $s_n = 17s_{n-2} - 73s_{n-3}$ for $n \geq 3$.
   c) $s_0 = 5$ and $s_n = s_{n-1}$ for $n \geq 1$.
   d) $s_0 = 3$, $s_1 = 5$ and $s_n = 2s_{n-1} - s_{n-2}$ for $n \geq 2$.

4) The number of bacteria in culture is 1000 (approximately), and this number increases 250% every two hours. Use a recurrence relation to determine the number of bacteria present after one day.

5) Find the solution to $a_n = 3a_{n-1} + a_{n-2} - 3a_{n-3}$ for n = 3, 4, 5, . . . , with $a_0 = 4$, $a_1 = 10$, and $a_2 = 12$.